

MS&E 228 (Winter 2026) — Lecture 7 Notes (Student Handout)

Covariate Adjustment in RCTs: When and How

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Readings. *Applied Causal Inference Powered by ML and AI*, §1-2. Relevant papers: Lin (2013), Negi & Wooldridge (2021), Guo et al. (2021).

1 Introduction: The Role of Covariates in Randomized Trials

In the previous chapter, we established the doubly robust (DR) estimator as the state-of-the-art method for estimating average treatment effects in observational studies, prized for both its robustness and its optimal precision. We concluded by examining its properties in the simplified setting of a Randomized Controlled Trial (RCT), where the propensity score is known and constant. This led to a surprising and powerful result: in an RCT, the DR estimator remains consistent for the ATE even if the outcome model is misspecified. More specifically, the DR estimator remains asymptotically normal and provides correct confidence intervals even if the outcome model does not converge to the true conditional expectation function, as long as it converges to *some* limit. This realization opens up a critical new line of inquiry: if we don't need to get the outcome model right to ensure consistency, what is the purpose of covariate adjustment in an experiment?

The answer is **variance reduction**. While randomization ensures that our estimate will be centered on the true ATE on average, it does not protect us from the vagaries of sampling variability. If our outcome of interest is noisy, our ATE estimate will also be noisy. By using pre-treatment covariates X to explain away predictable variation in the outcome, we can dramatically improve the precision of our estimate, allowing us to detect smaller effects with greater confidence and statistical power.

Key Question from Lecture 6

Is the variance of the DR estimator with a misspecified g_* better or worse than the simple two-means estimator?

Today, we investigate this question for a very common and practical choice of $\hat{g}$: the one estimated via simple Ordinary Least Squares (OLS). We will establish a surprising and fundamental equivalence between the OLS coefficient on the treatment variable and the DR estimator

in an RCT. Using this equivalence, we will perform a variance analysis and understand when OLS guarantees precision improvement.

2 OLS Recovers the ATE in Randomized Trials

Our first goal is to build a formal bridge between the world of doubly robust estimation and the familiar world of ordinary least squares and through that bridge argue that the OLS coefficient on D is always consistent for the ATE in RCTs. We will show that these two approaches, which seem conceptually distinct, are in fact deeply connected in the context of an RCT.

2.1 From OLS Minimization to the Normal Equations

Let's begin by recalling the mechanics of OLS. When we run a simple linear regression of the outcome Y on the treatment D and a set of covariates X , we are finding the coefficients $(\hat{\theta}, \hat{\beta}, \hat{c})$ that minimize the sum of squared residuals:

$$\min_{\theta, \beta, c} \mathbb{E}_n[(Y - \theta D - \beta' X - c)^2]$$

where $\mathbb{E}_n[\cdot]$ denotes the empirical average over our sample.

The solution to this minimization problem is characterized by the **normal equations** (the first order conditions to the optimization problem), which state that the regression residuals must be empirically orthogonal to (i.e., have zero covariance with) all the regressors in the model. Let $\hat{\epsilon} = Y - \hat{\theta}D - \hat{\beta}'X - \hat{c}$ be the OLS residuals. The empirical normal equations are:

$$\mathbb{E}_n[\hat{\epsilon} \cdot (1; D; X)] = 0 \quad (\text{empirical normal equations})$$

This single vector equation gives us three key conditions that the OLS residuals must satisfy:

1. $\mathbb{E}_n[\hat{\epsilon}] = 0$ (The residuals have a mean of zero.)
2. $\mathbb{E}_n[\hat{\epsilon} \cdot D] = 0$ (The residuals are uncorrelated with the treatment.)
3. $\mathbb{E}_n[\hat{\epsilon} \cdot X] = 0$ (The residuals are uncorrelated with the covariates.)

These orthogonality conditions are the defining mechanical property of OLS, and they are the key to unlocking the OLS-DR equivalence.

Remark: OLS finding solution to the Normal Equations

It is crucial to remember that OLS can be viewed as a method for solving the normal equations to find a linear model whose residuals are uncorrelated with the input features. This property holds even when the true conditional expectation function is not linear, and it will be heavily utilized throughout this and subsequent lectures.

2.2 Equivalence of OLS Coefficient and DR with OLS Outcome Regression

Now, let's see what happens when we use the linear model estimated by OLS as the outcome model, $\hat{g}$, in our DR formula (without cross-fitting). In an RCT with a known, constant treatment probability q , the DR estimator (without cross-fitting)¹ is:

$$\hat{\theta}_{\text{DR}} = \mathbb{E}_n \left[\hat{g}(1, X) - \hat{g}(0, X) + \frac{D - q}{q(1 - q)}(Y - \hat{g}(D, X)) \right]$$

Our model is $\hat{g}(D, X) = \hat{\theta}D + \hat{\beta}'X + \hat{c}$. Let's substitute this into the DR formula piece by piece.

Step 1: The G-formula Part. The first part of the DR formula is the average of the estimated CATE. For a linear model, the CATE is constant and equal to the coefficient on D :

$$\hat{g}(1, X) - \hat{g}(0, X) = (\hat{\theta}(1) + \hat{\beta}'X + \hat{c}) - (\hat{\theta}(0) + \hat{\beta}'X + \hat{c}) = \hat{\theta}$$

Therefore, the g-formula part of the estimator is simply $\mathbb{E}_n[\hat{\theta}] = \hat{\theta}$.

Step 2: The Correction Term. The second part of the DR formula is the IPW-style correction term. The residual in this term, $Y - \hat{g}(D, X)$, is exactly the OLS residual, $\hat{\epsilon}$. So the DR estimator becomes:

$$\hat{\theta}_{\text{DR}} = \hat{\theta} + \mathbb{E}_n \left[\frac{D - q}{q(1 - q)} \cdot \hat{\epsilon} \right] = \hat{\theta} + \frac{1}{q(1 - q)} \mathbb{E}_n[(D - q)\hat{\epsilon}]$$

Step 3: The Correction Term Vanishes! The final step is to show that the correction term is identically zero. We can expand the empirical average and use the normal equations:

$$\mathbb{E}_n[(D - q)\hat{\epsilon}] = \mathbb{E}_n[D\hat{\epsilon}] - q\mathbb{E}_n[\hat{\epsilon}] = 0 - q \cdot 0 = 0$$

This completes the proof. The de-biasing step of the DR estimator has no effect in this setting, as the OLS coefficient is already unbiased.

OLS Coefficient

When using a simple linear model $\hat{g}(D, X) = \hat{\theta}D + \hat{\beta}'X + \hat{c}$ for the outcome regression in an RCT, the doubly robust estimator (without crossfitting) is **numerically identical** to the OLS coefficient $\hat{\theta}$ on the treatment variable D .

2.3 A Powerful Consequence: OLS Consistency for the ATE

This equivalence leads to a powerful and practical result.

OLS Consistency for ATE in RCTs

The coefficient $\hat{\theta}$ on D in an OLS regression of $Y \sim D, X$ converges to the true ATE when applied to RCT data, **even when the true Conditional Expectation Function (CEF) of Y given D and X is not linear.**

¹Cross-fitting is not required in the DR estimator when using only parametric models for the outcome and the propensity; the asymptotic normality theorem is still valid. It is only required when using complex ML models.

The proof follows directly from the equivalence we just established: 1) We know the DR estimator converges to the ATE in an RCT (if $\hat{g}$ converges to *some* limit). 2) We just showed the DR estimator is identical to the OLS coefficient $\hat{\theta}$. Therefore, $\hat{\theta}$ must also converge to the ATE.

This is a non-trivial result. It tells us that even if our linear model is wrong, OLS still finds the right answer for the ATE in an experiment. The intuition is that randomization makes the treatment assignment D uncorrelated with the covariates X , which prevents the misspecification of the relationship between Y and X from biasing the estimate of the effect of D . This is a significant result because it implies that for RCTs, one can simply use a linear adjustment for covariates without needing to worry about correctly specifying the functional form of the outcome model.

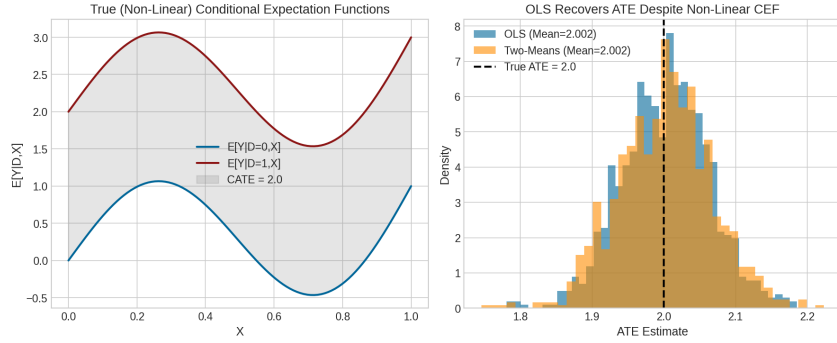


Figure 1: Even when the true CEF is non-linear (black curve), the OLS regression line (red) still yields a coefficient on D that correctly estimates the ATE. *DGP*: $X \sim N(0,1)$, $D \sim \text{Bernoulli}(0.5)$, $Y = 2D + \cos(2\pi X) + N(0,1)$. The true ATE is 2. The OLS estimate of the coefficient on D is also approximately 2. See Jupyter Notebook.

Discussion: Does The Equivalence of OLS and DR Hold Outside of RCTs?

A natural question is whether this property—that the correction term vanishes—holds outside of RCTs. The answer is that for the correction term to vanish, the residuals of OLS must be orthogonal to the inverse propensity weights $H_{p_0}(D, X)$. In an RCT, the propensity scores are constant, so this condition simplifies to the residuals being orthogonal to a constant and to D , which is guaranteed by the normal equations ($\mathbb{E}_n[\hat{\epsilon}] = 0$, $\mathbb{E}_n[\hat{\epsilon} \cdot D] = 0$).

In a general observational study, to make the correction term vanish empirically, one would need to include the inverse propensity weights as a feature in the OLS regression. This is known as the “**clever covariate**” approach, and it provides another way to mechanically compute the DR estimate using OLS. Specifically, if you include $\frac{D}{e(X)} - \frac{1-D}{1-e(X)}$ as a regressor, then the DR is equivalent to apply the g-formula to the resulting model. One way to operationalize this even more cleanly, is to add as part of the features X , the variables $1/e(X)$ and $1/(1-e(X))$ and then perform an interactive OLS regression $D, X, D \cdot (X - \bar{X})$. Then the coefficient on D in the resulting OLS regression will correspond to the g-formula applied to this model. Moreover, the residuals of this model will be orthogonal to the variables $D/e(X)$, $D/(1-e(X))$ and $1/(1-e(X))$. Thus the debiasing correction will be zero and the coefficient on D will equal to the DR estimator.

3 When Does OLS Adjustment Improve Precision?

We have established that OLS gives a consistent estimate of the ATE in an RCT. But does it help with precision? Does it reduce variance compared to the simple two-means estimator? The answer, surprisingly, is **not always**.

3.1 Setting Up the Variance Analysis

To analyze the variance, we look at the population analogues of the OLS problem.

Population Best Linear Predictor: Consider the linear model that minimizes the population prediction loss:

$$g_*(D, X) = \theta D + \beta' X + c \quad \text{where} \quad (\theta, \beta, c) = \arg \min \mathbb{E}[(Y - \theta D - \beta' X - c)^2]$$

Population Normal Equations: The solution satisfies the population normal equations:

$$\mathbb{E}[\underbrace{(Y - \theta D - \beta' X - c)}_{\epsilon} \cdot (1; D; X)] = 0$$

Let ϵ denote the **population residuals**. Since we argued that OLS is consistent, we also know that it must be that $\theta = \text{ATE}$.

3.2 Variance of the Linearly Adjusted Estimator

Since the DR estimator is equivalent to the OLS coefficient, we can use the DR variance formula. The asymptotic variance of $\sqrt{n}(\hat{\theta} - \theta)$ is:

$$V_{\text{OLS}} = \text{Var} \left(\theta + \frac{D - q}{q(1 - q)} \cdot \epsilon \right)$$

Because $\mathbb{E}[\epsilon] = 0$ and $\mathbb{E}[\epsilon \cdot D] = 0$ from the population normal equations, this simplifies to:

$$V_{\text{OLS}} = \frac{\mathbb{E}[(D - q)^2 \epsilon^2]}{(q(1 - q))^2}$$

Connection to Robust Standard Errors

This is exactly the formula for the asymptotic **heteroskedasticity-robust** standard errors for the parameter θ . Most OLS packages provide a finite sample approximation to this formula as the **heteroskedasticity-robust** standard errors **HC0**.^a It is a simplified version of the famous “**sandwich formula**”.

^atypically not the default standard errors in most OLS packages; but the OLS standard errors that you should always be using in this class.

Here is how you would implement this in Python:

```
import numpy as np
import statsmodels.api as sm

# 1. Fit the OLS model
```

```

X_reg = sm.add_constant(np.c_[D, X])
model = sm.OLS(Y, X_reg).fit()

# 2. The ATE estimate is the coefficient on the treatment D
# Note: Index is 1 because column 0 is the constant
ate_estimate = model.params[1]

# 3. Get heteroskedasticity-robust standard errors
# This is an estimate of the DR variance formula
ate_se = model.get_robustcov_results(cov_type='HCO').bse[1]

print(f"ATE Estimate: {ate_estimate:.4f}")
print(f"Robust S.E.: {ate_se:.4f}")

```

3.3 Variance of the Two-Means Estimator

The variance of the simple two-means estimator ($\bar{Y}_1 - \bar{Y}_0$) is:

$$\text{Var}(\text{Two-Means}) = \frac{\text{Var}(Y|D=1)}{q} + \frac{\text{Var}(Y|D=0)}{1-q}$$

Two-Means as OLS

The two-means estimator is equivalent to $\text{OLS}(Y \sim D)$, which learns a model $\hat{\theta}D + \hat{\theta}_0$. The coefficient $\hat{\theta}$ converges to ATE and $\hat{\theta}_0$ converges to $\mathbb{E}[Y | D=0] = \mathbb{E}[Y(0)]$.

Applying the equivalence of Two-Means to DR with outcome regression $g_*(D, X) = \theta D + \theta_0$, we can derive the variance formula:

$$V_{\text{TM}} = \frac{\mathbb{E}[(Y - \theta D - \theta_0)^2 (D - q)^2]}{(q(1 - q))^2}$$

This formula has the same structure as V_{OLS} , but the residuals are different.

3.4 Comparing the Variances

Let's relate the two residuals. Let $\epsilon_{\text{TM}} = Y - \theta D - \theta_0$ be the two-means residual. Then:

$$\epsilon_{\text{TM}} = (Y - \theta D - \beta'X - c) + (\beta'X + c - \theta_0) = \epsilon + (\beta'X + c - \theta_0)$$

where we crucially used that the coefficient θ on D is the same in both OLS specifications, since in either case it has to be the ATE. Plugging this into the variance formula for the two-means estimator and expanding:

$$\begin{aligned}
V_{\text{TM}} = & \underbrace{\frac{\mathbb{E}[\epsilon^2 (D - q)^2]}{(q(1 - q))^2}}_{V_{\text{OLS}}} + \underbrace{\frac{\mathbb{E}[(\beta'X + c - \theta_0)^2 (D - q)^2]}{(q(1 - q))^2}}_{\text{Positive Term}} \\
& + 2 \underbrace{\frac{\mathbb{E}[\epsilon(\beta'X + c - \theta_0)(D - q)^2]}{(q(1 - q))^2}}_{\text{Problematic Cross-Term}}
\end{aligned}$$

If the cross-term were zero, we would be done! V_{OLS} would always be smaller than V_{TM} .

3.5 The Problematic Cross-Term

The cross-term contains expressions like:

$$\beta' \mathbb{E}[\epsilon \cdot X \cdot D]$$

From the population normal equations for $OLS(Y \sim D, X)$, we know:

- $\mathbb{E}[\epsilon \cdot D] = 0$ ✓
- $\mathbb{E}[\epsilon \cdot X] = 0$ ✓
- $\mathbb{E}[\epsilon] = 0$ ✓

But...

We **do not** know that $\mathbb{E}[\epsilon \cdot X \cdot D] = 0$! This would only be true if the CEF were truly linear (i.e., $\mathbb{E}[\epsilon|D, X] = 0$). We only have the weaker “orthogonality” conditions that the OLS residuals satisfy.

3.6 When OLS Adjustment Can Hurt Precision

Because $\mathbb{E}[\epsilon \cdot X \cdot D]$ is not guaranteed to be zero, the cross-term can be substantially negative. This negative term can be large enough to overwhelm the positive term in the variance decomposition.

The Surprising Result

It is possible for $\text{Var}(\text{Two-Means}) < \text{Var}(\text{OLS Adjusted})$! Simple OLS adjustment is **not** guaranteed to improve precision.

This surprising result occurs because, in an unbalanced design, the covariates X can be correlated with the term $(D - q)^2$, which can create a problematic correlation in the cross-term. The issue is particularly pronounced when there is **treatment effect heterogeneity**—that is, when the effect of treatment varies with the covariates. When the design is balanced ($q = 1/2$), the cross-term vanishes, and OLS adjustment is always guaranteed to be at least as good as the two-means estimator.

Homework Preview

You will construct a simple data generating process where this exact scenario occurs. You will also prove that when $q = 1/2$ the problem vanishes and OLS is always weakly better than the two-means estimator.

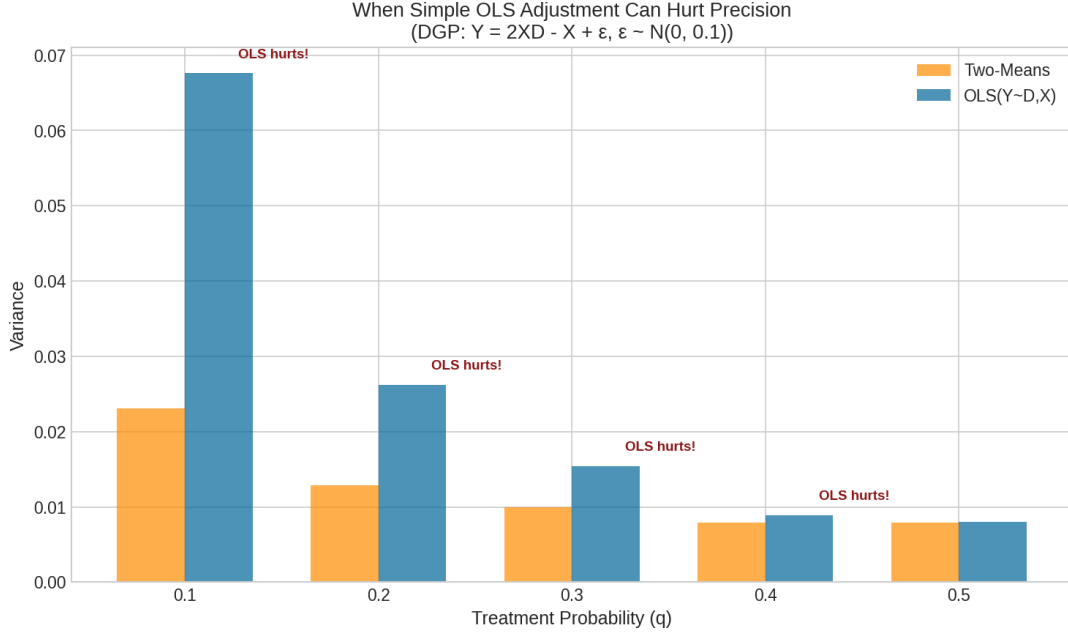


Figure 2: When the experiment is unbalanced ($q \neq 0.5$), simple OLS adjustment can increase variance compared to the two-means estimator. *DGP*: $X \sim N(0, 1)$, $D \sim \text{Bernoulli}(q)$, $Y = 2XD - X + \varepsilon$ where $\varepsilon \sim N(0, 0.1)$. The true ATE is $\mathbb{E}[2X] = 0$ since $\mathbb{E}[X] = 0$. The figure shows the variance of ATE estimates from 1000 simulations ($n = 500$ each) for different values of q . When $q \neq 0.5$, the OLS-adjusted estimator can have higher variance than the two-means estimator due to the interaction term creating a problematic cross-term in the variance decomposition. See Jupyter Notebook.

4 Interactive OLS for Guaranteed Improvement

How can we fix this problem and guarantee variance reduction? The issue arose because our OLS residuals were not orthogonal to the interaction term $D \cdot X$. The solution is to force this orthogonality by adding the interaction term to the regression itself.

Recommended Specification: The Interactive OLS Specification

For analyzing RCTs, it is advisable that you run OLS with interactions:

$$\text{OLS}(Y \sim D, X, D \cdot (X - \bar{X}))$$

where $\bar{X} = \mathbb{E}_n[X]$. This is sometimes called a “fully interacted model”. De-meaning the covariates is not so important but simplifies variance calculations.

This estimates a model:

$$\hat{g}(D, X) = \hat{a}_0 D + \hat{a}_1' (X - \bar{X}) \cdot D + \hat{\beta}' X + \hat{c}$$

CATE Estimate: The Conditional Average Treatment Effect is now a function of X :

$$\widehat{\text{CATE}}(X) = \hat{g}(1, X) - \hat{g}(0, X) = \hat{a}_0 + \hat{a}_1' (X - \bar{X})$$

ATE Estimate: We can estimate the ATE via the empirical g-formula (which is equivalent to the DR-formula, by the equivalence of OLS and DR in RCTs):

$$\hat{\theta} = \mathbb{E}_n[\widehat{\text{CATE}}(X)] = \hat{a}_0 + \hat{a}'_1(\mathbb{E}_n[X] - \bar{X}) = \hat{a}_0$$

Thus, our estimate is again the coefficient $\hat{a}_0$ on D (because we de-meaned the covariates before interacting them).

The new OLS residuals $\hat{\epsilon}$ will now satisfy the empirical normal equations:

$$\mathbb{E}_n[\hat{\epsilon} \cdot (1; D; X; D \cdot X)] = 0$$

This guarantees $\mathbb{E}_n[\hat{\epsilon} \cdot D \cdot X] = 0$.

4.1 Variance of the Interactive OLS Estimator

Since the debiasing term in the DR formula is still zero due to normal equations, we can analyze the variance of the DR score with the new limit predictor $g_*(D, X) = a_0D + a'_1(X - \mathbb{E}[X]) \cdot D + \beta'X + c$:

$$\mathbb{E} \left[\left(a_0 + a'_1(X - \mathbb{E}[X]) - \text{ATE} + \frac{D - q}{q(1 - q)}\epsilon \right)^2 \right]$$

After expanding the square and noting that by the orthogonality of the residuals with respect to $(1, D, X, D \cdot X)$ the cross-term vanishes, i.e.,

$$\mathbb{E} \left[(a_0 + a'_1(X - \mathbb{E}[X]) - \text{ATE}) \cdot \frac{D - q}{q(1 - q)}\epsilon \right] = 0$$

we can simplify the variance as:

$$\text{Var}(a_0 + a'_1(X - \mathbb{E}[X])) + \frac{\mathbb{E}[\epsilon^2(D - q)^2]}{(q(1 - q))^2}$$

The first term, $\text{Var}(a_0 + a'_1(X - \mathbb{E}[X]))$, can be thought as the variance of the best linear approximation of the CATE. The second term is the same form as before, but now with residuals ϵ that satisfy $\mathbb{E}[\epsilon \cdot D \cdot X] = 0$. This ensures this term is smaller than or equal to the variance of the two-means estimator. With this simple observation we can argue that the first term is smaller than the variance of the two-means. However, we are still left with the first term $\text{Var}(a_0 + a'_1(X - \mathbb{E}[X]))$. This would typically be a much smaller quantity. In fact, the recent work of Negi and Wooldridge, proves that the sum of both terms will always be weakly smaller than the variance of the two-means.

Guaranteed Precision Improvement (Lin 2013; Negi & Wooldridge 2021)

The overall asymptotic variance of the interactive OLS estimator is guaranteed to be less than or equal to the variance of the two-means estimator.

4.2 Practical Note: Estimating the Asymptotic Variance

The asymptotic variance has two components:

$$\underbrace{\text{Var}(a_0 + a_1'(X - \mathbb{E}[X]))}_{V_1} + \underbrace{\frac{\mathbb{E}[\epsilon^2(D - q)^2]}{(q(1 - q))^2}}_{V_2}$$

- The second term (V_2) is exactly what the HC0 variance $\hat{V}_2$ for $\hat{a}_0$ in OLS approximates, **if you de-mean the covariates X** before passing to OLS. Most statistical packages can compute this directly.
- Add to this variance the empirical variance of the estimated CATE model ($\hat{V}_1$).
- Use this variance when calculating the standard error (i.e., $\hat{se} = \sqrt{(\hat{V}_1 + \hat{V}_2)/n}$).

```
import numpy as np
import statsmodels.api as sm
import patsy

# De-mean covariates and create interaction terms
X_dm = X - X.mean(axis=0)
X_interact = patsy.dmatrix('D * (' + '+'.join(X_dm.columns) + ')',
                           X_dm.assign(D=D), return_type='dataframe')

# Run OLS with HC0 standard errors
ira = sm.OLS(Y, X_interact)
ira_results = ira.fit(cov_type='HC0')
ira_est = ira_results.params['D']

# Account for error in means of X (first-order effect on ATE)
interaction_cols = [c for c in X_interact.columns if c.startswith('D:')]
cate = X_dm @ ira_results.params[interaction_cols]
ira_se = np.sqrt(ira_results.HC0_se['D']**2 + np.var(cate)/len(Y))
```

Poll Question

In an unbalanced RCT, which OLS specification is *guaranteed* to have a variance that is at most the variance of the simple two-means estimator?

- (A) $\text{OLS}(Y \sim D, X)$
- (B) $\text{OLS}(Y \sim D, X, D \cdot X)$
- (C) $\text{OLS}(Y \sim D)$
- (D) Both (B) and (C)
- (E) Both (A) and (C)

Answer: The correct answer is (D). $\text{OLS}(Y \sim D)$ is the two-means estimator itself, so its variance equals that of the two-means estimator. $\text{OLS}(Y \sim D, X, D \cdot X)$ is the interactive model, which is guaranteed to have variance at most equal to the two-means estimator. The simple $\text{OLS}(Y \sim D, X)$ is *not* guaranteed to improve precision.

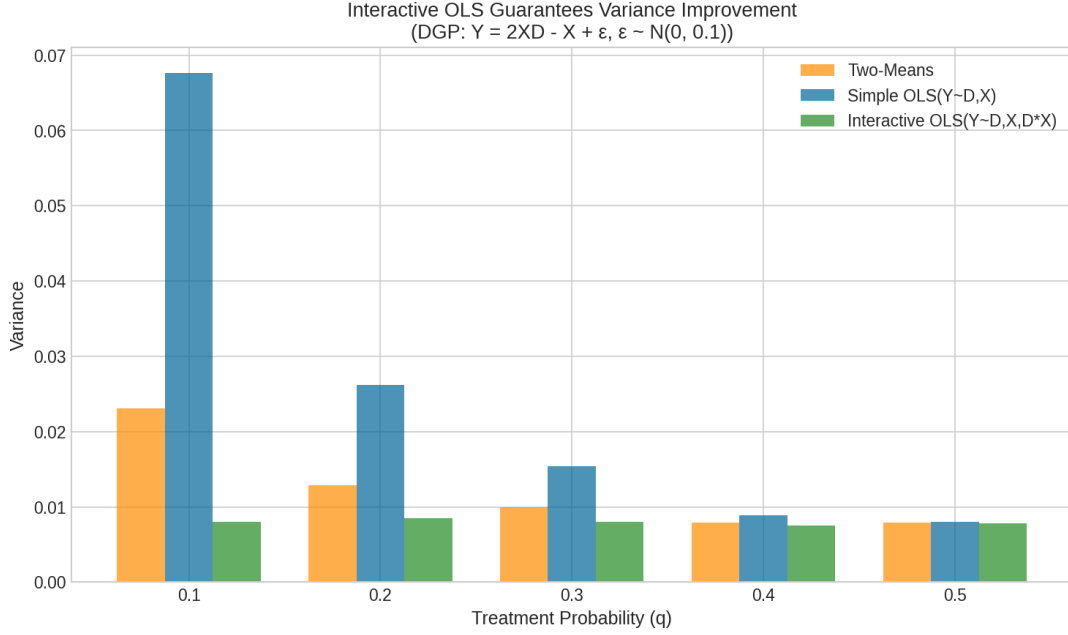


Figure 3: Interactive OLS guarantees variance reduction over the two-means estimator for any treatment probability q . DGP: $X \sim N(0,1)$, $D \sim \text{Bernoulli}(q)$, $Y = 2XD - X + \varepsilon$ where $\varepsilon \sim N(0,0.1)$. The true ATE is $\mathbb{E}[2X] = 0$ since $\mathbb{E}[X] = 0$. The figure shows the variance of ATE estimates from 1000 simulations ($n = 500$ each) for different values of q . The interactive OLS estimator (green) always has lower or equal variance compared to the two-means estimator (orange), regardless of the treatment probability q . In contrast, simple OLS (blue) can have higher variance than two-means when $q \neq 0.5$. See Jupyter Notebook.

4.3 Extension: ML-Learned Features (Guo et al., 2021)

We can go further by using ML-learned features $\hat{\phi}(X)$:

1. Train ML models (trees, neural nets) to predict Y from X (using cross-fitting).
2. Use their out-of-fold predictions $\Phi = \hat{\phi}(X)$ as additional features.
3. Run $\text{OLS}(Y \sim D + X + D \cdot (X - \bar{X}) + \Phi + D \cdot (\Phi - \bar{\Phi}))$.

The ATE estimate is again the coefficient a_0 associated with D . The variance can be calculated as in the previous Interactive Regression section. **This methodology was proposed recently by researcher's at Meta for variance reduction in A/B testing!**

Discussion: Connection to Stacking

Is this approach related to stacking. The answer is yes: the final OLS step is effectively performing a form of stacking by learning the optimal linear combination of the base learners' predictions, separately for the treated and the control. The ML models provide nonlinear features that capture complex patterns in the data, and the interactive OLS then optimally combines these features to maximize variance reduction.

5 Continuous Treatments

Finally, we switch back to general observational settings and briefly touch upon the challenge of continuous treatments. Here, the treatment D is not binary but can take on a range of values (e.g., drug dosage, price, investment amount). This represents a transition back from RCTs to observational studies, with an added layer of complexity.

Many real-world questions involve treatments that are not binary:

- **Healthcare:** What is the **dose-response curve**? How does patient response change with the dose of a drug?
- **Economics:** What is the **demand curve**? How does demand for a product change with its price?
- **Operations Management:** What is the **return on investment**? If we invest x dollars, what is our return per dollar?

Clarification: Continuous-Valued vs. Continuously-Administered Treatments

By continuous treatments we don't mean a treatment that persists over time, but simply a treatment that was administered once, but takes a value among a continuum of real values. We are still in a **cross-sectional setting**, observing different units at a single point in time, where each unit may have received a different level of a real-valued treatment. The case of a single unit receiving multiple treatments over time will be covered later in the course under the topic of dynamic treatment effects.

5.1 Continuous Treatments and Potential Outcomes

The potential outcomes framework extends naturally:

- **Potential Outcomes:** $Y(d)$, where the treatment level d is now a continuous variable. We have infinitely many potential outcomes for each individual.
- **Conditional Exogeneity:** The assumption is the same. For any level d :

$$Y(d) \perp\!\!\!\perp D \mid X$$

- **Overlap:** This assumption becomes much stronger. We need the density of the joint distribution of (D, X) to be non-zero at points (d, X) for any value d and for any X in the support. This is harder to satisfy.

5.2 Causal Estimands for Continuous Treatments

Our questions of interest also become richer:

- **Mean Counterfactual / Dose-Response Curve:**

$$\mu(d) = \mathbb{E}[Y(d)]$$

This is a curve representing the average outcome if we intervene and fix the value of the treatment at some treatment level d . Many times, we will be centering the dose response curve around its value at some baseline dosage level (without loss of generality 0):

$$\text{ATE}(d) = \mathbb{E}[Y(d) - Y(0)]$$

- **Marginal Effect at a point d :**

$$\text{ME}(d) = \frac{\partial}{\partial d'} \mathbb{E}[Y(d')] \Big|_{d'=d}$$

How much does the outcome change on average if we perturb the treatment around some level d .

- **Average Marginal Effect (AME):** The average of the marginal effects, taken over the *natural distribution of observed treatment values D* .

$$\text{AME} = \mathbb{E}[\text{ME}(D)]$$

Discussion: How Would You Approach This?

If phased with such a continuous treatment problem, how would you approach it? There are two types of paths to take:

1. **Discretization:** Discretize the continuous treatment into bins and run an analysis on the binned data. While this makes the problem more familiar, it imposes a step-wise functional form on the dose-response curve and loses information.
2. **OLS Regression:** Simply run an OLS regression of the outcome on the continuous treatment and covariates. This assumes a linear dose-response curve.

Both approaches involve trade-offs between flexibility and the assumptions required for identification. We will go down the second path and later try to generalize it to be more flexible and comparable in flexibility to the first approach.

5.3 Identification: G-Formula and Its Challenges

Identification via G-Formula: Under conditional exogeneity and overlap, we can still use the g-formula:

$$\mathbb{E}[Y(d)] = \mathbb{E}[\mathbb{E}[Y(d)|X]] = \mathbb{E}[\mathbb{E}[Y|D = d, X]] = \mathbb{E}[g(d, X)]$$

where $g(d, X) = \mathbb{E}[Y|D = d, X]$.

Challenge: IPW Fails

IPW-style estimators do not work in the same way. For a continuous treatment D , what is the analogue of $\Pr(D = d|X)$? It's a conditional density, $f_{D|X}(d|X)$. Even so, the IPW weight would be $\mathbf{1}\{D = d\}/f_{D|X}(d|X)$, which is zero almost everywhere. We need a different approach.

This highlights a fundamental statistical challenge of the continuous treatment setting: we are always asking the model to extrapolate or interpolate between observed treatment levels. Unlike the binary case where we observe both treated and untreated units, with continuous treatments we never observe the exact counterfactual at any specific treatment level.

5.4 A Simple Approach: The Linear Model

Assumption. Suppose the conditional expectation of the potential outcome is linear:

$$\mathbb{E}[Y(d)|X] = \theta \cdot d + \beta'X + c$$

Under this assumption, our causal estimands simplify dramatically:

- **Dose-Response:** $\mu(d) = \mathbb{E}[Y(d)] = \theta \cdot d + \beta'\mathbb{E}[X] + c = \theta \cdot d + c'$, for some constant c' .
- **ATE(d):** $\mathbb{E}[Y(d) - Y(0)] = \theta \cdot d$
- **Marginal Effect:** $\frac{\partial}{\partial d}\mathbb{E}[Y(d)] = \theta$ (constant)

Key Insight

All causal estimands boil down to estimating the single parameter θ .

5.5 Identification without Overlap under Linearity

For any point (d, x) in the support of the distribution, by Conditional Exogeneity:

$$\mathbb{E}[Y(d) | X = x] = \mathbb{E}[Y(d) | D = d, X = x] = \mathbb{E}[Y | D = d, X = x]$$

Thus by our assumption on the form of the mean counterfactual outcome:

$$\mathbb{E}[Y | D = d, X = x] = \theta \cdot d + \beta'x + c$$

Identification Strategy

If the best linear predictor of Y from D, X has a unique coefficient associated with D , then the causal estimands can be uniquely identified without the overlap assumption.

This is a powerful result: under linearity, the model allows for extrapolation, which means we can identify causal effects even at treatment levels that were not in the support of the observed data distribution. This is in stark contrast to the nonparametric setting, where overlap is essential.

Estimation Strategy

Simply run $\text{OLS}(Y \sim D, X)$.

One question that remains open is: when do we expect the population best linear predictor of Y from D, X to have a unique coefficient? This brings us to a deeper problem of trying to better understand and interpret what a coefficient in an OLS problem really means. We will endeavor down this path in the next lecture and address the following key questions.

Key Questions for Next Lecture

1. When does the population best linear predictor $Y \sim D, X$ have a unique solution with respect to the coefficient θ on D ?
2. How can we better interpret the coefficient θ in the best linear predictor?
3. What are the drivers of high-variance for the estimated parameter $\hat{\theta}$?
4. What is the analogue of “overlap” under linearity?

These questions will be addressed in the next lecture, where they will also lead us to solutions based on more advanced machine learning-based methods for continuous treatment effect estimation.

6 Summary: Four Key Takeaways

1. **OLS coefficient = ATE in RCTs:** The coefficient on D in $\text{OLS}(Y \sim D, X)$ is a consistent estimator for the ATE in an RCT, even if the CEF is not linear.
2. **Simple OLS can hurt precision:** For unbalanced experiments ($q \neq 1/2$), simple OLS adjustment can increase variance compared to the two-means estimator.
3. **Interactive OLS guarantees improvement:** The fully interacted model, $\text{OLS}(Y \sim D, X, D \cdot X)$, is guaranteed to (weakly) reduce variance.
4. **Continuous Treatments:** We need to make some modeling assumptions, as otherwise the problem is statistically daunting and the overlap assumption is quite restrictive.

7 References

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